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Emergency App

International Project

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1. Introduction

1.1 Problem Statement

Travelers in Thailand can quickly reach their limits in emergency or problem situations. Especially for tourists, it is often difficult to find the right contacts, emergency numbers, or medical facilities in stressful moments. In addition, communication can be challenging due to language barriers, as many travelers do not speak Thai and, conversely, sufficient English is not understood everywhere.

Particularly in medical emergencies, minor health issues, the loss of important documents, or general orientation difficulties, there is a risk that help cannot be organized quickly enough or only insufficiently. Existing information sources are often confusing, not specifically tailored to tourists, or too slow to use in a concrete situation.

For this reason, the main challenge of the project was to develop a mobile application that provides travelers in Thailand with fast, simple, and understandable support. The app is intended to offer clear guidance both in acute emergencies and in less urgent situations, while centrally providing important information. The core features implemented in the project are directly derived from this initial problem.

1.2 Objective of the Application

The objective of the developed application is to provide tourists in Thailand with quick and simple assistance in both emergency and everyday situations. The mobile web app is designed to help users quickly find the right information and contact options in stressful or unfamiliar situations. Especially in medical emergencies or urgent problems, the application aims to provide clear guidance and facilitate quick access to important emergency numbers.

In addition, the application is intended to assist users in less urgent situations, such as minor health issues, orientation difficulties, or the loss of important documents. Through a clear user interface structure and simple navigation, the app is designed to remain intuitive and easy to use even under time pressure.

Another important aspect of the application is overcoming language barriers. By supporting both English and Thai, communication between travelers and local institutions can be facilitated. Furthermore, the application allows users to store important personal information, such as insurance details or emergency contacts, so that these can be quickly accessed when needed.

2. Project Idea and Concept

2.1 Origin of the Idea

The idea for developing the application arose from considering how travelers in a foreign country could receive support more quickly and easily in unexpected situations. During the concept phase of the project, it became clear that although many information resources for tourists already exist, they are often spread across different websites or applications and are therefore difficult to access quickly in a specific situation.

Based on this observation, the idea emerged to develop an application that brings together important support functions in one central place and provides a clear and simple structure. The focus was on creating a solution that remains easy to understand and usable with as few steps as possible, even under time pressure.

Another aspect behind the idea was that travelers often do not know which local emergency numbers or contact points are relevant in a particular country. Therefore, the application was designed to provide targeted information and possible actions that are specifically tailored to the needs of tourists and help them quickly orient themselves in unfamiliar situations.

2.2 Core Function of the App

To enable users to quickly find the information they need, the application was intentionally designed with a very simple structure. The home screen of the app is divided into three main sections: **Emergency**, **No Emergency**, and **Important Details**. This structure ensures that users can immediately select the appropriate section depending on their situation.

The **Emergency** section is intended for urgent situations. Here, users can directly access important Thai emergency numbers, allowing them to quickly request help in critical situations. The feature is deliberately designed so that it can be used with as few interactions as possible.

The **No Emergency** section provides support for less urgent situations. These include, for example, minor health problems, orientation difficulties, or the loss of documents. In this section, users are guided step by step through simple options in order to receive appropriate recommendations or helpful information.

The **Important Details** section allows registered users to store important personal information and quickly access it when needed. This may include details such as the user's name, insurance information, or an emergency contact. In emergency situations, these stored details can help provide important information quickly.

3. Requirements Analysis and Planning

3.1 Requirements Prioritization

At the beginning of the development phase, the requirements for the application were systematically collected and prioritized. The goal was to first identify the most important features necessary for the basic use of the app. For this purpose, the **MoSCoW method** was used, in which requirements are categorized into **Must-Have, Should-Have, and Could-Have**.

The collaboration tool **Miro** was used to jointly plan and visualize the requirements. Various ideas and features were collected on a digital board and then prioritized according to their importance. Features defined as **Must-Have** formed the foundation for the first version of the application.

These core features included direct access to important emergency numbers, simple navigation between emergency and non-emergency situations, and the option to store important personal information in the user profile. This structured prioritization allowed the development process to be planned efficiently and ensured that the focus remained on the most essential functions.

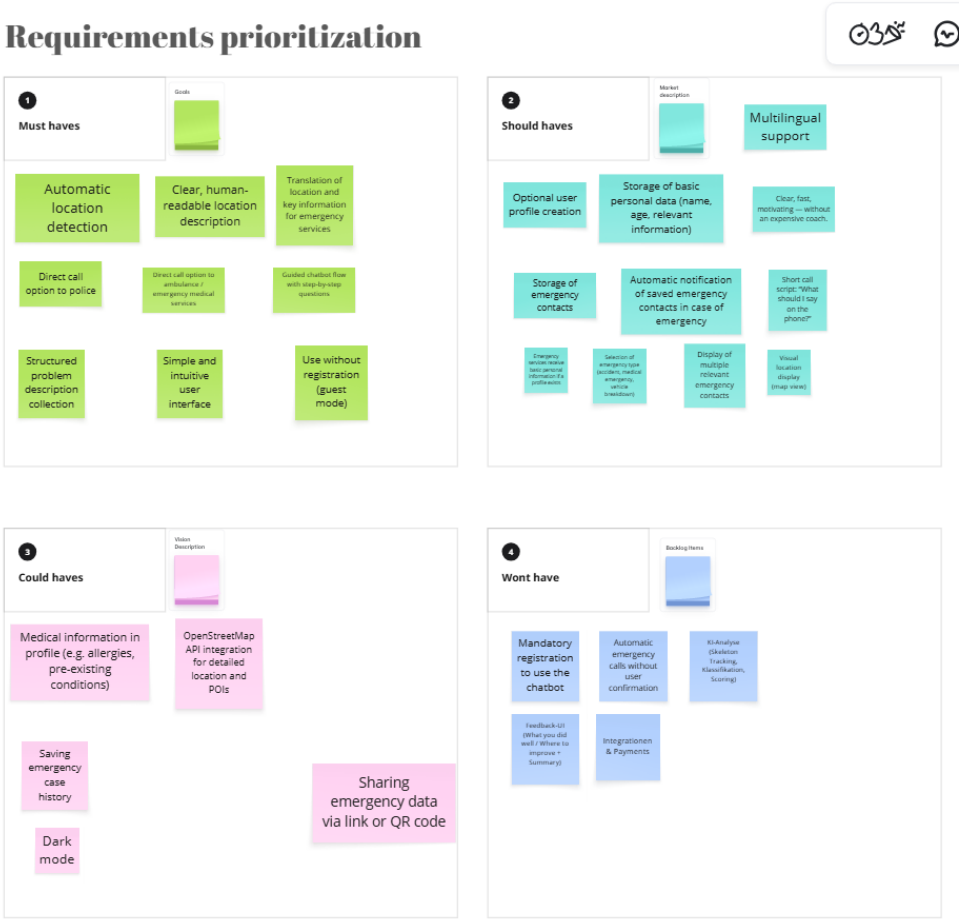

Abbildung 1: MoSCoW-Method

4. Design and Prototyping

4.1 Creation of the High-Fidelity-Prototype

After prioritizing the requirements, a visual prototype of the application was created. For this purpose, the design tool **Figma** was used to develop a **high-fidelity prototype**. This prototype closely represents the intended final appearance of the application and allows the design and navigation to be planned before the actual programming begins.

Within the prototype, the main screens of the application were designed, including the home screen and the different functional sections of the app. This made it possible to evaluate at an early stage whether the structure of the application was logically organized and whether the planned features could be presented in a clear and understandable way. The prototype subsequently served as the basis for the later technical implementation of the application.

4.2 User Interface Structure

When designing the user interface, particular emphasis was placed on a clear and simple structure. Since users may often find themselves in stressful or unfamiliar situations, the app was designed to be as intuitive as possible. For this reason, the home screen was structured so that the most important functions are directly visible and easily accessible.

The navigation within the application follows a simple structure in which users are guided step by step to the information they need. Large buttons, a clear arrangement of content, and a reduced number of options help make the app easy to understand and use. The goal was to ensure that users can quickly find the desired function without requiring extensive prior familiarization with the application.

4.3 Usability Considerations

Usability played a central role in the design of the application. Since the app is intended for international travelers, special attention was given to ensuring that the content is clearly understandable and written in simple language. In addition, the application was designed to be easily usable on smartphones as well as other mobile devices.

Another important aspect was reducing the number of interactions required. Users should be able to access important information or functions with as few clicks as possible. Furthermore, multilingual support was considered in order to reduce language barriers and make the application easier to use for international travelers. These considerations were intended to ensure that the application can be used quickly and reliably, even in stressful situations.

5. Technical Implementation

5.1 Frontend

The user interface of the application was implemented using **React**. React is a widely used JavaScript library for developing modern web applications and enables a component-based architecture. This allows individual parts of the user interface to be developed as reusable components, which improves the maintainability and scalability of the code.

Vite was used as the development environment. Vite provides a fast development server and short loading times during development, which helped make the development process more efficient.

For styling the application, **Tailwind CSS** was used. Tailwind is a utility-first CSS framework that allows layout and design to be implemented directly through predefined CSS classes. This made it possible to implement the visual design of the application quickly and consistently.

5.2 Navigation

For navigation within the application, **React Router** was used. This library enables the management of different pages within a **single-page application (SPA)**. As a result, users can switch between different views of the application without the need to reload the entire page.

By using React Router, various screens of the application—such as the home screen, the emergency section, and the profile area—could be logically connected. This ensures smooth navigation and improves the overall user experience within the application.

5.3 Backend and Database

Supabase was used for server-side functionality and the storage of user data. Supabase provides a **Backend-as-a-Service (BaaS)** solution and offers features such as authentication, database management, and API access.

Within the application, Supabase is primarily used for user registration and login as well as for storing profile information. This allows users to save personal data such as insurance details or emergency contacts and access them quickly when needed.

Using Supabase enables the rapid implementation of backend functionalities without the need to develop and maintain a separate server infrastructure.

5.4 Additional Technologies

In addition to the main technologies, several other tools and libraries were used to support specific functions of the application. For the display of icons and visual elements, the icon library **Lucide React** was used. This library provides a wide range of modern icons that can be easily integrated into React applications.

Furthermore, external interfaces are used to provide location-based information. This includes, for example, searching for nearby facilities such as hospitals, police stations, or pharmacies. These functions allow users to quickly find relevant services in their surroundings and help them navigate more easily in an unfamiliar environment.

6. Application Features

6.1 Emergency Function

One of the central features of the application is the **emergency mode**, which provides users with quick guidance in urgent situations. In this section, important Thai emergency numbers are clearly displayed, including the medical emergency number **1669** and the **tourist police number 1155**.

The feature is deliberately designed so that users can find the relevant contact options with as few steps as possible. The goal is to provide a fast and uncomplicated way to request help or access important information in stressful situations.

6.2 Support for Non-Urgent Situations

In addition to acute emergencies, the application also provides support for **non-urgent situations** that may occur during a trip. These include, for example, minor health issues, orientation difficulties, or general questions about what to do next.

The application guides users step by step through different options, allowing them to better identify their situation and receive appropriate recommendations. This feature is intended to reduce uncertainty and provide users with quick guidance.

6.3 Profile and Data Storage

Another component of the application is the ability to **store personal information in the user profile**. Registered users can enter details such as their nationality, date of birth, insurance information, or emergency contacts.

This data can be quickly accessed when needed and helps provide important information in critical situations. The information is stored using the integrated backend solution, ensuring that the data is managed securely.

6.4 Location-Based Assistance

The application also provides a **location-based search function** that allows users to find important facilities nearby. These include, for example, hospitals, police stations, and pharmacies.

By using the location feature, the app can display relevant facilities in the user's surroundings. This function is particularly helpful when users are in an unfamiliar environment and need to quickly find an appropriate place for assistance. As a result, navigation and orientation on-site are significantly improved.

7. Results and Evaluation

7.1 Achieved Goals

As part of the project, a functional **mobile web application** was developed that provides travelers in Thailand with quick assistance in various situations. The core features of the application were successfully implemented, including access to important emergency numbers, support for non-urgent problems, and the ability to store personal information within the user profile.

In addition, a clear and well-structured user interface was developed, enabling simple navigation between the different sections of the application. The previously defined core requirements were therefore largely fulfilled and provide a solid foundation for the further development of the application.

7.2 Strengths of the App

One of the main strengths of the application lies in its **clear and simple structure**. The most important functions are directly accessible from the home screen, allowing users to quickly reach the support they need. This is particularly beneficial in stressful situations.

Another advantage is the **multilingual functionality** of the application, which helps reduce language barriers. In addition, the app combines several useful features—such as emergency contacts, informational support, and the storage of personal data—within a single application. This creates a central platform that brings together various forms of assistance in one place.

7.3 Possible Improvements

Despite the successful implementation of the core features, there are several opportunities for further development of the application. For example, additional functions could be integrated, such as more precise navigation to nearby facilities or further informational resources for travelers.

Furthermore, the application could be expanded to include additional languages in order to reach a broader user base. A deeper integration of local information services or additional emergency-related information could also further increase the practical value of the app.

8. Conclusion and Outlook

8.1 Summary of the project

As part of the project, a **mobile web application** was developed to support travelers in Thailand in various situations. The goal was to create a clear and easily accessible solution that allows users to quickly find important information or contact options.

During the course of the project, the requirements were first defined and prioritized, followed by the creation of a design prototype and its subsequent technical implementation. The developed application combines several features, including access to emergency numbers, support for less urgent problems, the storage of personal information, and a location-based search for important facilities. This resulted in a functional foundation that can provide travelers with additional guidance and support.

8.2 Potential Further Development of the App

There are several possibilities for the future development of the application. Additional features could be integrated, such as a more detailed map integration or an expanded display of relevant facilities in the surrounding area.

Furthermore, extending the language support would be beneficial in order to make the application accessible to a broader group of users. The integration of additional information services for travelers could also increase the practical value of the app. Through such extensions, the application could be further improved in the long term and better adapted to the needs of international users.